

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To obtain the best fit line over single feature scattered datapoints using Linear Regression	
Experiment No: 02	Date:	Enrollment No: 92301733023

Aim: To obtain the best fit line over single feature scattered datapoints using Linear Regression

IDE: Google Colab **Theory:**

Linear regression is a method for determining the best linear relationship between two variables X and Y . If variables X and Y are uncorrelated, it is pointless embarking upon linear regression. However, if a reasonable degree of correlation exists between X and Y then linear regression may be a useful means to describe the relationship between the two variables. The usual approach is to use the *least-squares* method, which minimizes the squared difference between the actual data points and a straight line. Let $[x_i, y_i], i = 1, 2, 3, \dots, N$ be the N pairs of data values of the variables X and Y . The straight-line relating X and Y is $y = mx + c$, where m and c are the gradient and constant values (to be determined) defining the straight line. Thus, $y(x_i) - y_i$ is the difference between the line and data point i (see Fig. 1). Taking all the data points, we seek values of m and c that minimize the squared difference SD .

$$\sum_1^N [y(x_i) - y_i]^2$$

This is achieved by calculating the partial derivatives of SD with respect to m and c and finding the pair $[m, c]$ such that SD is at a minimum.

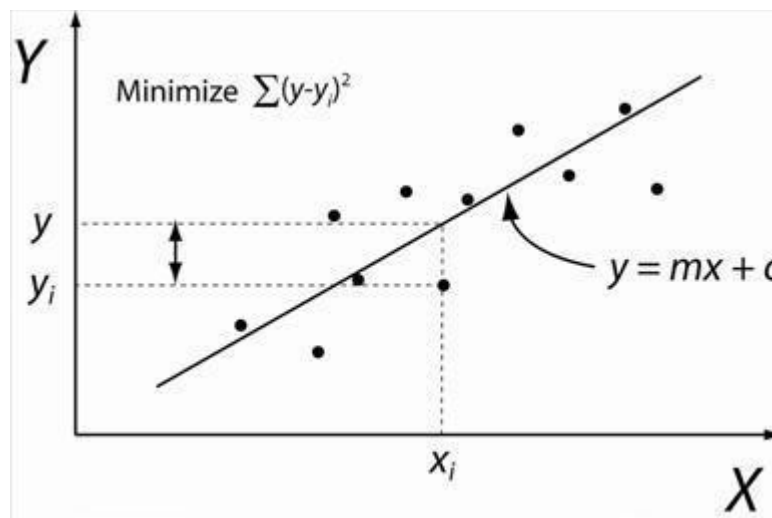


Figure 1: Illustration of Linear Regression. Linear least squares regression, the idea is to find the line $y = mx + c$ that minimizes the mean squared difference between the line and the data points

Batch Gradient Descent:

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Gradient Descent is an optimization algorithm used for minimizing the cost function in various machine learning algorithms. It is basically used for updating the parameters of the learning model. Batch gradient descent which processes all the training examples for each iteration of gradient descent. But if the number of training examples is large, then batch gradient descent is computationally very expensive.

Let m be the number of training examples. Let n be the number of features.

Algorithm for batch gradient descent :

Let $h_{\theta}(x)$ be the hypothesis for linear regression. Then, the cost function is given by:

Let Σ represents the sum of all training examples from $i=1$ to m . $J_{\text{train}}(\theta) = (1/2m) \Sigma (h_{\theta}(x^{(i)}) - y^{(i)})^2$

Repeat {

$\theta_j = \theta_j - (\text{learning rate}/m) * \Sigma (h_{\theta}(x^{(i)}) - y^{(i)})x_j^{(i)}$

For every $j = 0 \dots n$

}

Where $x_j^{(i)}$ Represents the j^{th} feature of the i^{th} training example. So if m is very large (e.g. 5 million training samples), then it takes hours or even days to converge to the global minimum. That's why for large datasets, it is not recommended to use batch gradient descent as it slows down the learning.

Pre Lab Exercise:

a. Explain the meaning of linear regression

Linear Regression is a statistical and machine learning method used to find the relationship between two variables (X and Y) by fitting a straight line equation $y = mx + c$. It predicts the output variable (Y) based on the input variable (X) by minimizing the error between actual and predicted values using least squares method.

b. Write three applications of linear regression

1. Predicting student marks based on study hours
2. House price prediction based on area
3. Sales forecasting in business

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c. Write three advantages of linear regression

1. Simple and easy to understand
2. Fast computation
3. Works well with small datasets

D. Write three limitations of linear regression

1. Assumes linear relationship
2. Sensitive to outliers
3. Cannot model complex relationships

Methodology:

1. Load the basic libraries and packages
2. Load the dataset
3. Analyse the dataset
4. Pre-process the data
5. Visualize the Data
6. Separate the feature and prediction value columns
7. Write the Hypothesis Function
8. Write the Cost Function
9. Write the Gradient Descent optimization algorithm
10. Apply the training over the dataset to minimize the loss
11. Find the best fit line to the given dataset
12. Observe the cost function vs iterations learning curve

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Program (Code):

```

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

data = {
    'Study_Hours': [2,5,3,6,4],
    'Marks': [55,78,50,88,70]
}

df = pd.DataFrame(data)

plt.scatter(df['Study_Hours'], df['Marks'])
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Scatter Plot")
plt.show()

X = df['Study_Hours'].values
Y = df['Marks'].values

m = 0
c = 0
learning_rate = 0.01
epochs = 1000
n = len(X)

cost_list = []

for i in range(epochs):
    Y_pred = m * X + c

    cost = (1/n) * sum((Y - Y_pred)**2)
    cost_list.append(cost)

```

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```
dm = (-2/n) * sum(X * (Y - Y_pred))
dc = (-2/n) * sum(Y - Y_pred)
```

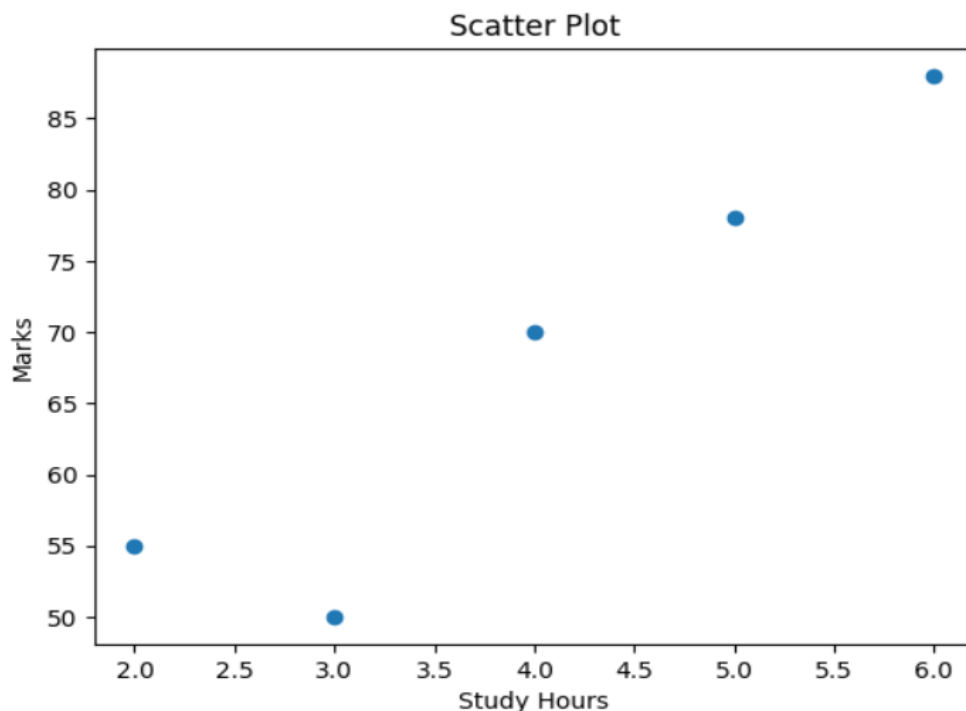
```
m = m - learning_rate * dm
c = c - learning_rate * dc
```

```
print("Slope (m):", m)
print("Intercept (c):", c)
```

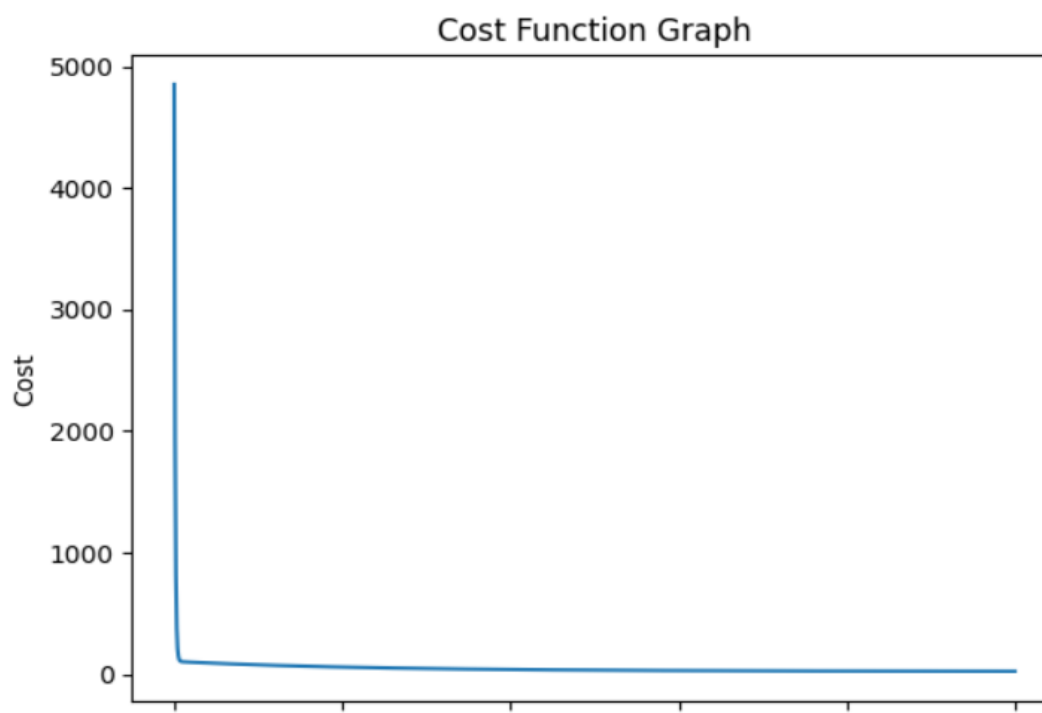
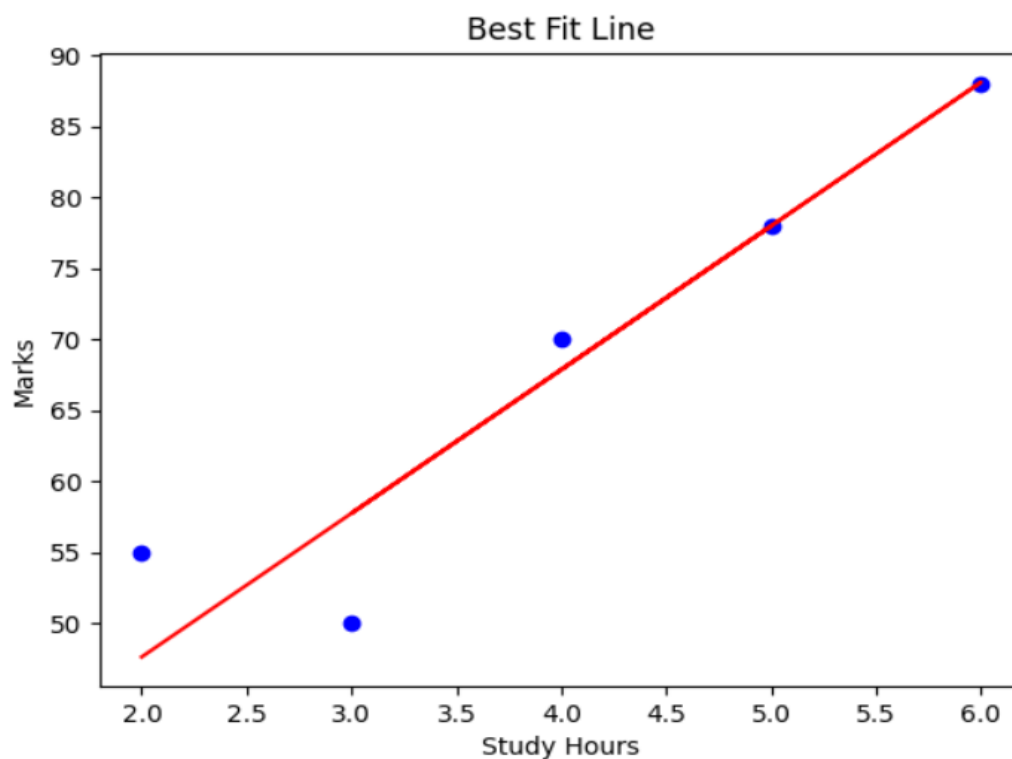
```
plt.scatter(X, Y, color='blue')
plt.plot(X, m*X + c, color='red')
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Best Fit Line")
plt.show()
```

```
plt.plot(range(epochs), cost_list)
plt.xlabel("Iterations")
plt.ylabel("Cost")
plt.title("Cost Function Graph")
plt.show()
```

Results:



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Observation and Result Analysis:

a. Nature of the dataset

The dataset consists of numerical values representing **Study Hours (input feature)** and **Marks (output variable)**.

From the scatter plot, it is observed that as study hours increase, marks also increase, indicating a **positive linear relationship** between the variables.

The data is simple, small in size, and suitable for applying linear regression.

b. During Training Process

During training using Gradient Descent:

- The model starts with initial values of slope ($m = 0$) and intercept ($c = 0$).
- In each iteration, the predicted values are calculated and compared with actual values.
- The **cost function (Mean Squared Error)** gradually decreases.
- Parameters (m and c) are updated continuously using gradients.
- The learning process moves towards minimizing the error and finding the optimal line.

c. After the training Process

After completing all iterations:

- The algorithm converges to optimal values of slope (m) and intercept (c).
- A **best-fit line** is obtained that closely represents the data points.
- The prediction accuracy improves significantly.
- The cost function reaches a minimum value, indicating successful training.

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Post Lab Exercise:

- a. What are the major assumptions considered in linear regression
 1. There is a **linear relationship** between independent and dependent variables.
 2. The residuals (errors) are **normally distributed**.
 3. The variance of errors is **constant (homoscedasticity)**.
 4. The observations are **independent** of each other.
 5. There is **no multicollinearity** (for multiple features).
- b. Why MSE is used instead of MAE for calculating the loss function
 1. MSE squares the errors, so **larger errors are penalized more**, improving accuracy.
 2. It is **differentiable**, making it suitable for Gradient Descent optimization.
 3. It provides a smooth curve, which helps in faster convergence compared to MAE.
- c. How can the behaviour of outliers be understood while dealing with the unseen dataset
 1. Outliers significantly affect the regression line because they increase the error.
 2. The model may shift towards outliers, reducing overall accuracy.
 3. Outliers can be detected using visualization (scatter plot) or statistical methods like **IQR or Z-score**.
 4. Handling outliers (removal or transformation) improves model performance.
- d. Derive the Normal Equation for the Linear Regression.

The Normal Equation is used to directly compute the best parameters without iteration.

$$\theta = (X^T X)^{-1} X^T Y$$

Where:

- X = Feature matrix

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- Y = Output vector
- θ = Parameters (m and c)

It minimizes the cost function and gives the optimal solution directly.